

Investigating Simple Harmonic Motion

What is a torsional oscillator?

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30 March 1977

Resonance

From the resonance peak (see Figure), the resonant angular driving frequency, ω_D , is $(3.26 \pm 0.02) \text{ rad s}^{-1}$.

Using the resonance shift equation (5), we get a value for the natural frequency of $(3.28 \pm 0.01) \text{ rad s}^{-1}$. This is lower than the predicted value from free oscillations $[(3.33 \pm 0.01) \text{ rad s}^{-1}]$.

Amplitude of oscillation against driving frequency of force at 0.6A

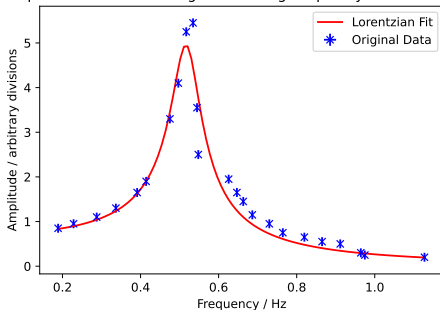


Figure: A resonance curve with Lorentzian fit for a damping current of 0.6 A. The peak falls at $(0.519 \pm 0.004) \text{ Hz}$.